CA3 & Hopfield networks

(discrete attractors)

Computational Neuroscience

University of Bristol

Adapted from C O'Donnell

mrule

Learning outcomes:

- Be able to explain what the Hopfield energy function is,
  - how it relates to stored patterns, and
  - how it evolves under asynchronous updates
- Hebbian plasticity and the Hopfield network

Hopfield network

Recurrent Neural Network (RNN)

- Discrete time (Δ steps, not $\mathrm{d}/\mathrm{d}t$)
- Discrete states $x_{i;t} \in \{-1, 1\}$
- McCulloch-Pitts (linear threshold) units

Proposed by John Hopfield (1982)

- Basic model of associative memory recall
- ***2024 Nobel Prize in Physics***

States evolve dynamically, typically toward some "attractor" state. A synaptic plasticity rule can imprint attractors in the network weights.

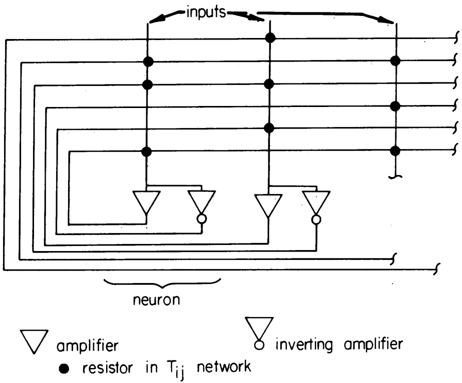


FIG. 2. An electrical circuit that corresponds to Eq. 5 when the amplifiers are fast. The input capacitance and resistances are not drawn. A particularly simple special case can have all positive T_{ij} of the same strength and no negative T_{ij} and replaces the array of negative wires with a single negative feedback amplifier sending a common output to each "neuron."

Hopfield (1984) PNAS

Hopfield networks

Weights (in this course)

- Always symmetric, $e_{ij} = w_{ji}$
- No self edges (no autapses), $W_{ii} = 0$

State dynamics:

- Synchronous:

$$\boldsymbol{x}_{t+1} \leftarrow \operatorname{sign}[\mathbf{W}\boldsymbol{x}_t - \boldsymbol{\vartheta}]$$

- Asynchronous: Pick a unit index i, then

$$x_{i;t+1} \leftarrow \operatorname{sign}[\textstyle\sum_{ij} w_{ij} x_{t;j} - \vartheta_i]$$

- States evolve to minimize an "energy":

$$E = -\frac{1}{2}\sum_{ij} w_{ij} x_i x_j + \sum_i \vartheta_i x_i$$
$$= -\frac{1}{2}\boldsymbol{x}^\top \mathbf{W}\boldsymbol{x} + \boldsymbol{\vartheta}^\top \boldsymbol{x}$$

Today we'll arbitrary declare $\operatorname{sign}(0) = -1$

no autapses!, but you might find self-weights in past exams

was that clear?

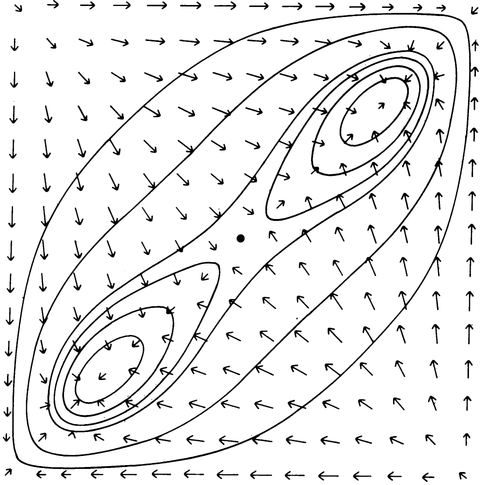


FIG. 3. An energy contour map for a two-neuron, two-stable-state system. The ordinate and abscissa are the outputs of the two neurons. Stable states are located near the lower left and upper right corners, and unstable extrema at the other two corners. The arrows show the motion of the state from Eq. **5**. This motion is not in general perpendicular to the energy contours. The system parameters are $T_{12} = T_{21} = 1$, $\lambda = 1.4$, and $g(u) = (2/\pi)\tan^{-1}(\pi\lambda u/2)$. Energy contours are 0.449, 0.156, 0.017, −0.003, −0.023, and −0.041.

Hopfield, PNAS 1984

Energy 'Landscape'

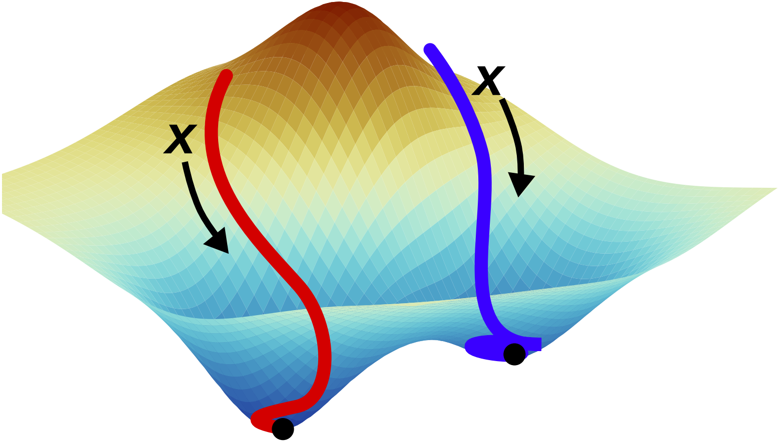
Local minima → 'attractors'

The network can do pattern completion: retrieving the full pattern from a partial cue.

The network we study here has capacity (maximum # stored attractors) of $\sim 0.14N$.

do we need to be be more specific with the math definitions?

Pattern completion dynamics ~ descending energy landscape



was that clear?

Learning attractors in Hopfield networks

Consider a Hopfield network with N units (neurons)

We have a set $\mathcal{X}$ of P patterns $\boldsymbol{x}_a = \{x_{1;a}, x_{2;a}, \ldots, x_{N;a}\}^\top$ (column vectors), where

- each $x_{i;a} \in \{-1, 1\}$, and
- $a \in 1 \ldots P$ indexes over patterns
- $i \in 1 \ldots N$ indexes over neurons

Define zero-mean patterns

$$\tilde{x}_{i;a} = x_{i;a} - \underbrace{\tfrac{1}{N} \textstyle\sum_{i=1}^{N} x_{i;a}}_{\mu_a}$$

$$\tilde{\boldsymbol{x}}_a = \boldsymbol{x}_a - \mu_a$$

Store desired attractor states into the synaptic weights using a learning rule:

$$w_{ij} = \begin{cases} \frac{1}{P} \sum_{a=1}^{\ldots P} \tilde{x}_{i;a} \tilde{x}_{j;a} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases} \tag{1}$$

$$\mathbf{W} = \langle \tilde{\boldsymbol{x}} \tilde{\boldsymbol{x}}^\top \rangle_{\mathcal{X}} - \mathbf{I}$$

Is the matrix/vector/expectation notation too cryptic?

write this with/without vector notation, write this as code

was that clear?

Is this really a biologicla learning rule?

Surprisingly ok model of CA3 *Symmetric STDP*!

Hebbian plasticity

When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."—Donald Hebb (1949)

"neurons that fire together wire together"—Dr. Carla Shatz (c. 1992)

To a first approximation:

$$\boxed{\Delta w_{ij;t} \propto x_{i;t} x_{j;t} \qquad (i \neq j)}$$

Verify (for $\vartheta = 0$), scaling of weights $c\mathbf{W}$ by a positive contant does not change the dynamics, so we can present patterns sequentially $t := a$ and use a variant of Equation (1) that is quite similar to Hebbian learning:

$$\mathbf{W}_{t+1} \leftarrow \mathbf{W}_t + (\tilde{\boldsymbol{x}}_t - \mu_t)(\tilde{\boldsymbol{x}}_t - \mu_t)^\top$$

end

page left blank for notes

page left blank for notes